# Tulip

Agastya Nath 12340140
Paritosh Lahre 12341550
Rahul Dev Reddy 12342390

Network Science
Indian Institute of Technology Bhilai

February 10, 2026



# Overview

- Open-source graph visualization and analysis framework
- Focus on large networks (up to $\approx 10^6$ nodes)
- Technical Stack:
  - C++ core for performance
  - Qt GUI for user interface
  - Python scripting for automation
- **Domains:** Social, biological, software, and communication networks
- **Users:** Researchers, analysts, and students

# Why Tulip in Network Science?

- Designed for very large graphs
- Interactive visual exploration of complex structure
- Many built-in layouts and metrics
- Plugin system for custom algorithms

# Architecture (High-Level)

- C++ core framework
- Graph data model with node/edge properties and hierarchies
- Plugin-based algorithms and import/export

# Backend Details

- Property-based graph storage
- Support for nested subgraphs / clusters
- Filtering and inheritance mechanisms
- OpenGL rendering engine

# Frontend and Extensibility

- Multiple linked views (graph, matrix, histograms, scatter plots)
- Rich styling: color, size, shape, labels from properties
- C++/Python plugins for layouts, metrics, importers
- Python scripting interface for automation

# Input / Output Formats

- Native: TLP (text, binary, JSON)
- Common imports: GML, CSV
- Export: TLP, images/screenshots, property tables
- Good interoperability with other graph tools

# Supported Network Types

- Directed and undirected graphs
- Weighted networks via properties
- Hierarchical / clustered graphs
- Large static networks ($\approx 10^6$ nodes, $\approx 5 \times 10^6$ edges, hardware-dependent)

# Handling Temporal / Multilayer

- Time / layer encoded as node/edge properties
- Filters for specific time ranges or layers
- No dedicated dynamic animation engine (workarounds only)

# Core Features: Construction

- Import from TLP, GML, CSV
- Manual editing inside GUI
- Programmatic construction via C++ / Python
- Flexible attribute types (numeric, string, categorical)

# Core Features: Metrics

- Degree, betweenness, closeness, etc. (via plugins)
- Clustering / community detection algorithms
- Path-based and connectivity measures
- Distribution plots of properties (histograms)

# Core Features: Layouts and Visualization

- Force-directed and stress layouts
- Access to OGDF layouts through plugins
- Node size/color mapped to properties
- Multiple coordinated views (graph + plots)

# Core Features: Scripting

- Python console inside Tulip
- Batch pipelines for import $\rightarrow$ layout $\rightarrow$ metrics $\rightarrow$ export
- Easy prototyping of new algorithms
- Integration with external Python ecosystem (e.g., pandas, NumPy)

# Scalability and Performance

- Optimized in-memory graph representation
- Efficient rendering for large graphs via OpenGL
- Handles $\approx 10^6$ nodes, $\approx 5 \times 10^6$ edges in typical scenarios
- Some algorithms can use multi-threading (plugin-dependent)

# Strengths

- Strong focus on large-scale graph visualization
- Rich set of built-in layouts and metrics
- Highly extensible via plugins and Python scripts
- Multiple, coordinated visual views

# Limitations

- Steep learning curve for new users
- Visualization-centric, not a full ML/statistical suite
- Limited native support for animated dynamic networks
- Performance depends strongly on hardware

# Use Case 1 (Social Network)

- Input: small social / collaboration graph
- Apply force-directed layout
- Color by community, size by degree
- Observe hubs and community structure

# Use Case 2 (Protein Network)

- Protein-protein interaction network
- Run centrality algorithms
- Apply force-directed layout
- Color by functional modules
- Size by degree / betweenness
- Identify hubs & modules

# Conclusion

- Tulip = scalable framework for large graph visualization and analysis
- Best when networks are big and structure is complex
- Strengths: layouts, metrics, extensibility, visual analytics
- Good choice for research-oriented network science projects

# References


- David Auber, *Tulip: An information visualization software for huge graphs*, Graph Drawing (GD 2003), Springer, 2003.
- Wikipedia contributors, *Tulip (software)*, `https://en.wikipedia.org/wiki/Tulip_(software)`, 2025.
- Tulip Developers, *Tulip: Large graphs analysis, drawing and visualization framework*, `https://github.com/Tulip-Dev/tulip`, 2025.
- Tulip Project, *Tulip: Data visualization framework*, `https://tulip.labri.fr`, 2025, LaBRI, University of Bordeaux.